Exercise 1.F: From Design to Implementation

How do we implement the fixed Part E architecture as real retrieval code?

Part F implements the same multimedia retrieval architecture twice over Flickr8k.

Classical MDM / classical ML pipeline
explicit descriptors
required k-NN over handcrafted visual features

Modern deep-learning pipeline
CLIP ViT-B/32 learned embeddings
FAISS IndexFlatIP retrieval

Validation note
FAST_DEV runtime validation passed on a 200-image subset.
The full Flickr8k run is explicitly deferred.
No final benchmark conclusion is drawn from these slides.

representation choice

indexing choice

query processing

evaluation design

Why Flickr8k Fits Exercise 1.F

What is the multimodal object?

Multimedia object
one image
five captions
metadata

Structured store
image id, width, height, aspect ratio

Semi-structured store
five caption fields per image
treated as a list of text fields

Media/content DB
JPEG image content

Text query -> retrieve images

Image query -> retrieve visually similar images

Text + image query -> fused retrieval

Bridge to Part E: Flickr8k instantiates the fixed Part E architecture using real multimedia objects.

Reference: content/lab/1E/slide-E-draft-ali_merge 2.pdf

Pipeline 1: Classical MDM / Classical ML Pipeline

How can we retrieve images without deep learning?

Pipeline 1 is a classical MDM / classical ML pipeline: explicit representations plus k-NN over handcrafted features. It excludes deep learning, not machine learning.

Text path
concatenate five captions
compute TF-IDF vectors
rank by cosine similarity

Visual path
extract HSV color histogram
extract Canny/edge-density descriptor
classical ML: required k-NN over handcrafted descriptors

Fusion path
combine text and visual scores
only for real dual text+image queries

Forbidden in Pipeline 1
CLIP, SigLIP, BERT, CNN/transformer encoders, neural embeddings/classifiers, depression-detection components
VNN, OCR, NRClex, Empath, XGBoost

The baseline is deliberately interpretable and instance-based for the visual retrieval component.

Pipeline 1 Mapped to the Fixed Part E Architecture

Where does the classical ML code sit in the architecture?

Offline ingestion

Flickr8k
images + captions

Metadata stores
structured + caption lists

Feature extraction
TF-IDF descriptors
HSV histogram
Canny/edge-density

Feature/descriptor DB
TF-IDF matrix +
handcrafted visual matrix

Online query path

Query descriptor extraction
text query -> TF-IDF vector
image query -> HSV/Canny descriptor

Query processor / query engine
TF-IDF cosine ranking
visual k-NN search
late score fusion

Results / visualization
ranked thumbnails
scores

Stores
media/content DB
structured store
semi-structured store

Canny/edge-density is a nested low-level visual step, not a top-level architecture module.

Pipeline 2: Modern Deep-Learning CLIP + FAISS

What changes in the modern deep-learning pipeline?

Pipeline 2 is the modern deep-learning pipeline: CLIP ViT-B/32 learns image/text embeddings; FAISS retrieves vectors.

Image encoder
L2-normalized image embedding

Text encoder
L2-normalized text embedding

openai/clip-vit-base-patch32
shared image-text embedding space

FAISS IndexFlatIP
main index
inner product over normalized vectors = cosine

Ranked results
thumbnails + scores

Optional scalability discussion
IVF/nprobe may be discussed as an approximate-search extension; not the main Flickr8k setting.

Guardrails
SigLIP2 is background/optional only. ViT patching inside CLIP is not the Part E image-segmentation module.

Pipeline 2 Mapped to the Fixed Part E Architecture

How does the modern deep-learning pipeline still follow Part E?

Offline ingestion

Flickr8k
images + captions

Metadata stores
structured + caption lists

Feature extraction
CLIP image embeddings
cache under
content/lab/1F/outputs/

Feature/descriptor DB
CLIP image embedding matrix

Online query path

Query descriptor extraction
text query -> CLIP text embedding
image query -> CLIP image embedding

Query processor / query engine
FAISS IndexFlatIP search

Results / visualization
ranked thumbnails
scores

Stores
media/content DB
structured store
semi-structured store

Key point: The architecture is reused; the representation changes.

What Stayed Constant, What Changed, and How We Evaluate

What makes the comparison fair?

Constant across both pipelines
same dataset: Flickr8k
same query file: evaluation_queries.json
same three evaluation paths
same metrics: Precision@K, Recall@K

Evaluation paths
evaluate_text_only(...)
evaluate_visual_only(...)
evaluate_fused(...)
alpha only in fused retrieval

Evaluation contract
required caption terms define relevance
optional terms do not define binary relevance
TF-IDF similarity is not ground truth
reference image excluded

Important limitation
visual/fused query anchors currently use placeholder reference ids with deterministic fallback.
Replace them with real Flickr8k filenames before treating visual/fused metrics as final.

Takeaway: Expected Trade-offs, Not Final Benchmark Claims

What should students learn?

Classical ML over explicit descriptors
explicit and inspectable descriptors
interpretable similarity logic
direct connection to classic MM indexing
less semantic generalization by construction

Deep learning over learned multimodal embeddings
image-text semantic embedding space
less manual feature engineering
vector-index-based retrieval
lower interpretability and model-dependent behavior

No final comparison verdict here
FAST_DEV validates the implementation workflow only. Full Flickr8k benchmarking is deferred.

Part F implements the fixed Part E retrieval architecture twice: explicit representations + classical ML, and CLIP deep-learning encoder + FAISS retrieval.